

Connectivity-Constrained Multi-Robot Exploration with Adaptive Formation Control

ROB 516: Advanced Multi-Robot Systems — Final Project

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Winter 2026

Motivation & Problem Statement

Why this matters

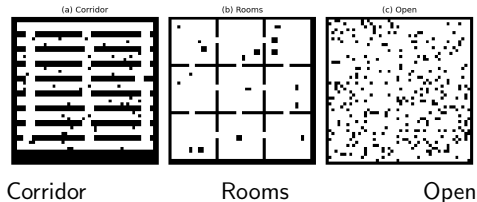
- Autonomous exploration: search & rescue, disaster response, planetary mapping
- Robot teams must *share maps* and *coordinate* wirelessly
- Connectivity $\Leftrightarrow \lambda_2(L) > 0$

The tension

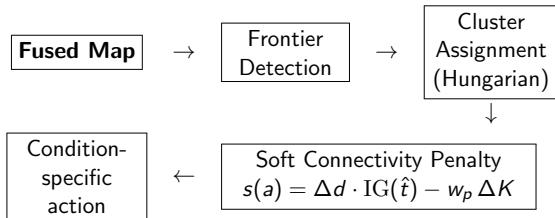
- Frontier exploration maximises coverage — robots spread without bound
- Hard connectivity veto eliminates disconnections — **coverage collapses** to $\leq 32\%$
- **Goal:** a smooth coverage–connectivity Pareto trade-off

Experimental setting

4 robots · 50×50 grid · 3 map types · 45 runs (5 conds
× 3 maps × 3 seeds)



System Pipeline — Overview



Five conditions compared across all three map types:

#	Condition	Core idea
1	Pure Frontier	Unconstrained frontier exploration (baseline)
2	Frontier + Conn	Frontier + soft connectivity penalty
3	Rigid Formation	Group convoy to same frontier cluster
4	Adaptive Formation	Per-robot frontier + cohesion pull
5	Adversarial	Condition 4 with one defecting robot

Metrics: final coverage (%), steps to 90%, disconnection events (steps with $\lambda_2 = 0$)

Condition 1 — Pure Frontier Exploration

Algorithm

- 1 Detect frontier cells (explored $\leftrightarrow$ unknown boundary)
- 2 Cluster frontiers by PCA-based region grouping
- 3 Assign robots via Hungarian algorithm:

$$U(i, j) = \text{IG}(t_j) - c_{\text{cost}} \cdot d(\mathbf{p}_i, t_j)$$

- 4 Plan paths with A^* — **no connectivity constraint**



Pure Frontier — corridor (steps 1→500)

Results across maps

	Corridor	Rooms	Open
Coverage	94.1%	98.8%	99.8%
Steps to 90%	159	128	112
Disconnections	483	479	137

Coverage ceiling — but robots freely scatter in corridors.

Condition 2 — Frontier with Soft Connectivity Penalty

Soft penalty replaces the hard connectivity veto:

$$s(a) = \underbrace{\Delta d \cdot \text{IG}(\hat{t})}_{\text{frontier progress}} - \underbrace{w_p \cdot \Delta K(a)}_{\text{connectivity cost}}$$

- ΔK = change in connected components after move a
- Disconnect only if frontier $> w_p$ unknown cells (information worth it)
- Greedy joint evaluation prevents simultaneous splits



Frontier+Conn — corridor (steps 1→500)

Penalty weight sweep — corridor map

w_p	Coverage	Disconnections
0 (baseline)	94.1%	483
5	93.1%	290
10	93.0%	164
25	90.1%	480
50	26.5%	0

$w_p = 10$: **66% fewer disconnections** at only 1.1 pp coverage cost

Condition 3 — Rigid Formation (Group Convoy)

Algorithm

- All robots navigate *together* to the highest-IG cluster
- Each robot assigned a distinct cell within the cluster
- **Multi-pass filter**: iterates assignment until no robot can be further unblocked — resolves relay-chain ordering deadlocks
- **Deadlock detection**: if all robots stay for $\tau = 15$ steps $\Rightarrow$ skip to next reachable cluster



Rigid Formation — corridor (steps 1→500)

Results

	Corridor	Rooms	Open
Coverage	50.6%	63.9%	82.8%
Disconnections	0	0	0

Only zero-disconnection method — at the cost of serial convoy parallelism.

Condition 4 — Adaptive Formation with Cohesion Control

Per-robot, per-step decision:

- 1 Individual frontier action (Hungarian, as Cond. 1)
- 2 Cohesion action — move toward group centroid
- 3 If $\text{drift} > r_{\text{eff}}$, override with cohesion:

$$r_{\text{eff}} = r_{\text{coh}}(1 - \alpha) = 20 \times 0.65 = 13 \text{ cells}$$

- 4 Apply soft penalty ($w_p = 10$) as final layer

Cohesion radius

$r_{\text{coh}} = 2.0 r_{\text{comm}} = 20 \rightarrow r_{\text{eff}} = 13 \text{ cells}$ (allows relay-chain spread across map)

Observed trade-off (corridor):

- Disconnections: 381 vs. 164 for Cond. 2
- Cohesion overrides fire *before* connectivity scorer
- Coverage: 71.2% (cohesion limits dead-end exploration)



Condition 5 — Adversarial Formation

Setup

- Identical to Condition 4, except:
- Robot $n-1$ **maximises** its Chebyshev distance from centroid each step
- Models a faulty or compromised team member
- Soft penalty still applies to adversary (no cohesion pull)

Coverage impact

	Corridor	Rooms	Open
Baseline (Cond. 1)	94.1%	98.8%	99.8%
Adversarial (Cond. 5)	66.6%	46.4%	73.7%
Drop	28 pp	52 pp	26 pp

- **Rooms** hit hardest: adversary pulls team out of rooms
- **Open** most resilient: wide space, others continue exploring
- Disconnections near-identical to unconstrained (484, 482, 477)



Adversarial — rooms (steps 1→500)

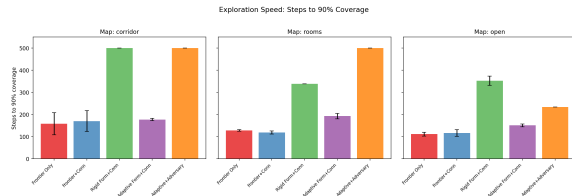
Results — Coverage and Exploration Speed

Final Coverage (%)

Condition	Corr.	Rooms	Open
1. Pure Frontier	94.1	98.8	99.8
2. Frontier+Conn	93.0	98.3	99.9
3. Rigid	50.6	63.9	82.8
4. Adaptive	71.2	96.6	98.4
5. Adversarial	66.6	46.4	73.7

Key takeaways

- Cond. 2 within 1–2 pp of unconstrained on all maps
- Cond. 4: near-full on rooms/open; 71% in corridor (cohesion limits dead ends)
- Rigid pays serial-convoy tax (−37 pp in corridor)
- One adversary destroys rooms coverage (−52 pp)



Steps to 90% (absent bar = never reached in 500 steps)

- Cond. 1 & 2 fastest: 110–175 steps across all maps
- Cond. 4 trails 20–65 steps; 1 of 3 corridor seeds never reaches 90%
- Cond. 3: never reaches 90% coverage — final coverage is 50.6%, 63.9%, and 82.8% on corridor, rooms, and open maps respectively

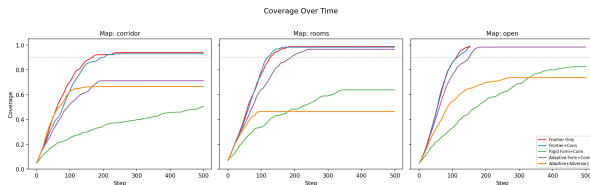
Results — Connectivity Analysis & Discussion

Mean Disconnection Events

Condition	Corr.	Rooms	Open
1. Pure Frontier	483	479	137
2. Frontier+Conn	164	475	123
3. Rigid	0	0	0
4. Adaptive	381	463	347
5. Adversarial	484	482	477

Three key insights

- 1 **Soft penalty works:** -66% disconnections at only 1.1 pp coverage cost (corridor)
- 2 **Map topology is a hard constraint:** rooms separation $> r_{\text{comm}}$ makes disconnections geometrically unavoidable
- 3 **Cohesion hurts connectivity:** Cond. 4 (381) worse than Cond. 2 (164) — cohesion fires before connectivity scorer



Coverage vs. step — all conditions (mean $\pm$ std, 3 seeds)

Pareto frontier (corridor): Cond. 2 dominates — near-maximum coverage *and* fewest disconnections among non-rigid conditions. Cond. 3 is the only zero-disconnection option (at coverage cost).

Conclusion & Future Work

Three algorithmic contributions

① IG-weighted soft penalty

$$s = \Delta d \cdot \text{IG} - w_p \Delta K$$

Coverage: $\leq 32\% \rightarrow \geq 93\%$; Disconnections: -66%

② Multi-pass connectivity filter

Resolves relay-chain ordering deadlocks in rigid formation

③ Corrected cohesion radius ($r_{\text{eff}} = 13$ cells)

Enables relay-chain spread without centroid clustering

Key findings

- Frontier+Conn is the best single-objective method
- Room separation $> r_{\text{comm}}$: geometric hard constraint no algorithm alone resolves
- Adaptive formation worthwhile only when formation structure itself is required

Limitations

- Chebyshev model ignores wall occlusion
- Greedy per-robot ordering — not globally optimal
- Discrete grid, simulation only

Future Work

- Control Barrier Functions for continuous-space connectivity
- Topology-aware (line-of-sight) communication model
- Joint action optimisation (MILP or learned policy)
- Real hardware validation with WiFi robots
- Heterogeneous teams with dedicated relay robots